PREPAID WATER CREDIT SYSTEM

Group C Evening

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INTRODUCTION

This system is a C-based console application designed to automate water tracking, manage user accounts, and handle prepaid token billing cleanly.

SYSTEM ARCHITECTURE

- **Uses Functions:** The program is broken down into small, separate tasks so it is easy to read and fix.
- **Array of Structures:** All information is stored in a simple list in the computer's memory for quick access.
- **50-Household Limit:** The database size is fixed to ensure the program runs fast without crashing.

THE HOUSEHOLD DATA BLUE PRINT

- **All-in-One Variable:** We created a custom struct to tie a user's ID, Name, Account Type, Balance, and Usage together.
- **Account Type Flag:** A simple number identifies whether the user is Domestic or Commercial.
- **The Blueprint:**

```
typedef struct {
    int id;
    char headName[50];
    char type[15]; // "Domestic" or "Commercial"
    float balanceUnits;
    float dailyUsageUnits;
} Household;
```

SYSTEM SECURITY

- **Login Gatekeeper:** A basic login function checks the user before showing the main menu.
- **Master PIN:** The system is protected by a simple access pin.
- **Pass or Fail:** If the password is right, it returns 1 to unlock the app; otherwise, it closes down.

THE MAIN MENU

- **Continuous Loop:** The program keeps running inside a loop so the admin can do multiple tasks without restarting.
- **Menu Routing:** A switch-case block takes the user's choice (1 to 7) and jumps directly to that specific task.
- **Wrong Input Guard:** If someone types a wrong number or letter, the program catches it and asks them to try again.

ADDING AND SEARCHING RECORDS

- **Registering Users:** Before adding a new household, the code checks if the list is full or if the ID already exists.
- **Searching by ID:** A basic for loop goes through the list one by one to find the matching ID.
- **Not Found Warning:** If the loop finishes and finds nothing, it safely tells the admin the record doesn't exist.

UPDATING AND DELETING RECORDS

- **Editing Info:** Once an ID is matched, the admin can directly type over the old name or update the daily water usage.
- **Deleting a Record:** To remove someone, the program locates their spot and slides all the remaining names up by one index to fill the gap.
- **Updating the Count:** The system automatically lowers the total user count so it doesn't leave blank spaces in the memory.

SAFETY CHECKS AND LIMITS

- **Low Credit Alert:** The code constantly checks if a user's balance drops below **5.0 units** to give a warning message.
- **Leak Detection:** If the system notices daily usage jump past **50.0 units**, it prints a critical alert for a possible broken pipe.
- **Payment Safeguard:** If a user deposits less money than the required maintenance fee, the code stops the math immediately and shows an error.

CONCLUSION

- This system provides a reliable, secure, and automated way to manage water consumption and prevent resource waste. By using a simple C array of structures to handle basic database tasks smoothly without heavy external software, the application integrates system controls like administrative PIN verification and automated warnings for data security, while its modular code structure allows for easy future upgrades such as connecting actual flow sensors or adding mobile payment integration.
- **Source Code:** https://github.com/Ellybenon/Prepaid-Water-Credit-System